

Module: CMP6046B Ubiquitous Computing

Assignment: Final Project

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Date set: 17 March 2026
Value: 55%

Date due: 21 May 2026 15:00
Returned by: 26 June 2026
Submission: Report and video submission on Blackboard
Checked by: Dr Xu Zhang

Learning outcomes

To gain experience in developing a multifaceted ubiquitous computing system from scratch that is designed to overcome a real-world problem identified by the student(s). To develop the ability to build their ubiquitous computing system to a proof-of-concept level and demonstrate feasibility.

Specification

Overview

This project is to be done in pairs unless otherwise stated.

To develop the proof-of-concept ubiquitous computing system based on your own idea that solves an existing problem in any setting. The system should include three elements **Sensing**, **Interpretation** and **Feedback**.

These elements do not necessarily need to be fully integrated and developing each element separately is permitted.

Description

Sensing and Data Collection

Using either supplied, purchased, students' own equipment or a combination, the students must collect and generate real world or mock data that is relevant to the problem. The students should identify the sensor types needed and if needed build the sensing system. A list supplied sensors will be made available.

A data collection protocol should be designed that either collects data directly from a real-world scenario or a close approximation. If particular outcomes of interest (labels) are also needed this and other supplementary information must also be collected in a robust way.

If your data collection requires acquiring sensed data from other people you must obtain consent from all participants. Please contact Dr Aung for advice on compiling consent documents for your specific project. Stored datasets must always be anonymised and be secured with at least one level of security (password and/or encryption). Data must also be anonymous and not contain any Personal Identifiable Information (PII).

Data Processing and/or Interpretation

The project should include a data interpretation stage which processes, utilizes and if needed make inferences on the collected data. The students should identify what data processing and analysis methods will be needed and also justify their use. The students must also implement, evaluate and demonstrate their effectiveness using appropriate metrics. **This stage doesn't not need to be fully integrated into the sensing apparatus and can be done separately if necessary.**

System Output and Feedback

The students must also include an appropriate design choice for the way processed data is given as feedback to the user or user(s). This can either be built in as an output device in an integrated way, an offline visualization or if needed or a lo-fi mock up if the other options are not possible. Usability testing with user trials is not expected but the output choices, designs and modalities should be justified.

If the students want to conduct user trials as part of this stage this is permitted but consent must be taken from all participants. Please contact Dr Aung for advice on compiling consent documents for your specific project. Survey questions must not ask private and confidential information or retain any information that is PII. Saved information must be anonymised and be secured with at least one level of security (password and/or encryption).

Relationship to formative assessment

This final assignment builds upon all prior formative and summative work as well as all lectures and supplementary materials.

Deliverables

1. **Presentation Video** - The students will give a 10-minute presentation in groups within a recorded video. The presentation will describe all aspects of the system and must include: problem statement, description of sensors and data collection, description of data processing and interpretation, description of feedback and output and any testing results. If feasible, a live demonstration of the project is should be included within the video. Please note there is a file size limit on BlackBoard, please be careful to ensure your video file does not exceed this limit, there is no restriction on video file format, but it must be accessible using Windows, mp4 is recommended.
2. **Report** – The students will write and submit via BlackBoard a joint report, the report should contain the following sections:
 - (i) **An agreed statement** from all group members declaring each student's % contribution to the project, the default is 50% for a group of 2.
 - (ii) **Aims, Motivation & Background section** (500 words approx.) – This section should describe the real-world problem the system aims to overcome. This section should also include descriptions of any other systems or solutions that exist that are similar and discuss their strengths and weaknesses. References and links to further information should be included if necessary.

(iii) **Sensing Section** (600 words approx.) – This section should describe the sensors, the reason for these choices and the general platform of the system (e.g. Ras Pi, Arduino or Phone) and sensor system structure. The section should describe what information gain/value each sensor gives in relation to the overall aim of the project and their strengths and weaknesses. It should also describe possible other alternative sensing approaches that could have been used for the same purpose.

(iv) **Data Collection Section** (400 words approx.) - This section should describe how data has been collected, if it is from a real-world scenario, mock scenario or an approximation. There should be justification of the collection method and protocols used with strengths and weaknesses discussed. This section should also give a summary description of the final dataset collected.

(v) **Data Processing/Interpretation Section** (600 words approx.) – This section should describe how the raw collected data has been processed in terms of data cleaning, feature engineering and use of modelling (e.g. machine learning), labelling and domain knowledge where appropriate. It should also describe how it has been evaluated and the output of the processing tested for reliability and accuracy.

(vi) **System Output and Feedback Section.** (500 words approx.) This section should describe the choice and design of all system output that are feedback to the user(s) and any lo-fi diagrams of images. This section should give reasoning for these choices and discuss the strengths and weaknesses. If any user testing was done, this should be discussed here.

(vii) **Conclusion Section** (300 words approx.) This section should give a concluding overview of the whole system. It should summarize the systems merits and shortcomings in relation to any testing/evaluation that was done in each of the sections as well as in relation to the original aims and the real-world problem to be solved.

The report should include images, figures, results tables, equations and diagrams where needed. Code, data tables, references and other listings should be placed as appendices. There is no restriction text font and document processing however the report should be clear and legible and cover page must include group members names, group name/number and the group members student ID numbers. The report should be saved as a PDF file.

NOTE: The word lengths are guidelines, some projects may require more or less information and description in some of the sections depending on the nature of the project. The key requirement is to describe all important aspects of the project in a proportionate and appropriate way. The whole report should approximate to 3000 words.

Resources

Main Materials

All hardware requirements will be dependent on the specific project idea. Students could draw from devices and components that exist within the Module's equipment

stock list (if able to return to the lab, please speak to Dr Aung if this is needed) or students can also supply own equipment subject to approval.

All software requirements will be dependent on specific project idea. The principal software in this module being Android Studio, Arduino IDE and Python (Anaconda) are available on the University network but are also free to download and install separately. All prior assignments and lecture materials support this final project.

Marking scheme

Activity	Marks	Awarded
Demo Video	20	
Report - Aims, Background & Conclusions	9	
Report - Sensing & Data Collection	13	
Report – Processing & Output/Feedback	13	
	Total	/55

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- what is meant by the terms 'plagiarism', 'collusion', and 'contract cheating'
- how to avoid plagiarism, collusion, and contract cheating
- using a proof reader
- what will happen if we suspect that you have breached the policy.

It is essential that you read this policy and you undertake (or refresh your memory of) our school's training on this. You can find the policy and related guidance here: <https://my.uea.ac.uk/departments/learning-and-teaching/students/academic-cycle/regulations-and-discipline/plagiarism-awareness>

The policy allows us to make some rules specific to this assessment. Note that:

- In this assessment, you are permitted to work within groups proscribed by the assessment setter. Discussing solutions to tasks *between* groups, or groups otherwise working together to perform the assessed tasks will be considered as a breach of university regulations. Please pay careful attention to the definitions of contract cheating, plagiarism and collusion in the policy and ask your module organiser if you are unsure about anything.